# Analysing the changes in bird distributions among the three Danish breeding bird atlases

The first part of chapter could be centered around five connected questions ( or we can pick those of them we like). They will tell us how the recorded distribution of Denmark's breeding birds changed, whether the balance between rare and common species changed, where (in which grid cells) communities changed most, and whether different parts of Denmark became more similar in their bird assemblages.

The range-size frequency analysis, which I suggested at our first meeting, is a useful addition. It gives us a national view that is easy to explain biologically. However, we should always read it alongside the geographically labelled records. A distribution can keep the same shape while the species occupying its different positions change, and a species can keep the same range size while moving to different cells.

The organising principle is to be precise about the object being compared. Sometimes it is a species–cell record, sometimes a named species, sometimes one cell through time, and sometimes two different cells within one atlas. These comparisons answer different questions. Their answers can differ without contradicting one another.

Throughout this letter, Atlas 1, Atlas 2 and Atlas 3 denote the three periods. All numerical examples and original figures are hypothetical or simulated (by chatgbt and only serves to illustrate things and give you an idea of what I mean). They are illustrative examples, not results for Denmark! We will leave climate and land-use explanations for the lsecond and third part chapter, which you have already done.

| Scientific question | Primary summary | Role in the chapter |
|---|---|---|
| 1. Complete occurrence pattern | Persistence, gains, losses; H and J | How much the labelled records changed. |
| 2. Danish range sizes | RSFD, $W_1$ and named-species changes | Whether the balance of restricted and widespread species changed. |
| 3. Systematic geographical change | Atlas and space × atlas partition | How much change follows the selected temporal and spatial structure. |
| 4. Exceptionally changed cells | Temporal Jaccard and conditional-null departures | Where local change is concentrated beyond a stated expectation. |
| 5. Homogenisation or heterogenisation | Change in mean spatial Jaccard | Whether cells' species lists became more or less alike. |

## I use four different meanings of "different"

**Description** asks how much the recorded matrices differ. If they cover our entire defined study system, the calculated difference is an observed property, not an estimate that automatically needs a standard error.

**Uncertainty estimation** needs a target. A spatial bootstrap can describe variation under a model of comparable spatial realisations. It cannot reconstruct missed detections. A detection model would address a different uncertainty.

**Hypothesis testing** asks whether a specified statistic is unusual under a specified model. "No change", "random placement of observed changes" and "no geographical interaction" are different null hypotheses.

**Practical similarity** asks whether change is smaller than a biologically chosen tolerance. A large p-value does not establish this. For example, an agreed tolerance of two percentage points in an occupancy summary would need a suitable interval lying within that tolerance. Two percentage points is an illustration, not a recommended universal threshold.

# Q1. How much did the complete occurrence pattern change?

## Biological question and quantities

One could start with the most direct question: how many species–cell records changed their status? A skylark occurrence in one cell and a skylark occurrence in another are different entries, even if their national total is unchanged.

For each pair of atlases, count four outcomes. Let a be persistent presences, b losses from presence to absence, c gains from absence to presence, and d persistent absences. There are N × S entries, so a + b + c + d = NS.

$$H_{tu} = \frac{b + c}{NS}, \qquad J_{tu} = \frac{b + c}{a + b + c}, \qquad \Delta R_{tu} = c - b.$$

H is the fraction of all entries that changed. J is whole-matrix Jaccard dissimilarity: the fraction that changed among species–cell combinations recorded in at least one of the two atlases. Net change is c − b records. H and J both range from zero to one, but answer different denominator questions.

Persistent absences enter H's denominator but not J's. In a sparse matrix, H can therefore look small despite considerable redistribution of recorded occurrences. Report both, together with the four counts. Neither number alone tells us whether the change consisted predominantly of gains, losses or balanced replacement of occurrences.

## Recommended analysis and software

These are direct calculations, implemented in the appendix with base R logical comparisons and sums. Look at Legendre (2019), in my opinion the best spatiacal ecological statisticians around, provides a methodological reference for temporal dissimilarity and gain/loss components at individual sites. Our pooling across the complete labelled matrix is a descriptive adaptation; its site-level significance test does not become a national atlas test by pooling entries.

Do not apply a binomial test treating NS entries as independent trials. Neighbouring cells and species share information, and a census of recorded entries is not a random sample of independent Bernoulli changes. Inference about systematic change belongs in question 3; survey uncertainty requires the information discussed above.

## Cookbook (I asked chatgbt to generate on how to do the analyses I suggest)

1. Take the three aligned binary matrices and confirm identical row and column names.
2. Calculate a, b, c and d for 1→2, 2→3 and 1→3. Confirm that their sum equals NS and that c − b equals the difference in total recorded presences.
3. Calculate H and J using the denominators above. Keep counts as well as proportions.
4. Repeat the gain/loss accounting by species and by cell. Identify whether a national total is concentrated in a few widespread species or particular regions.

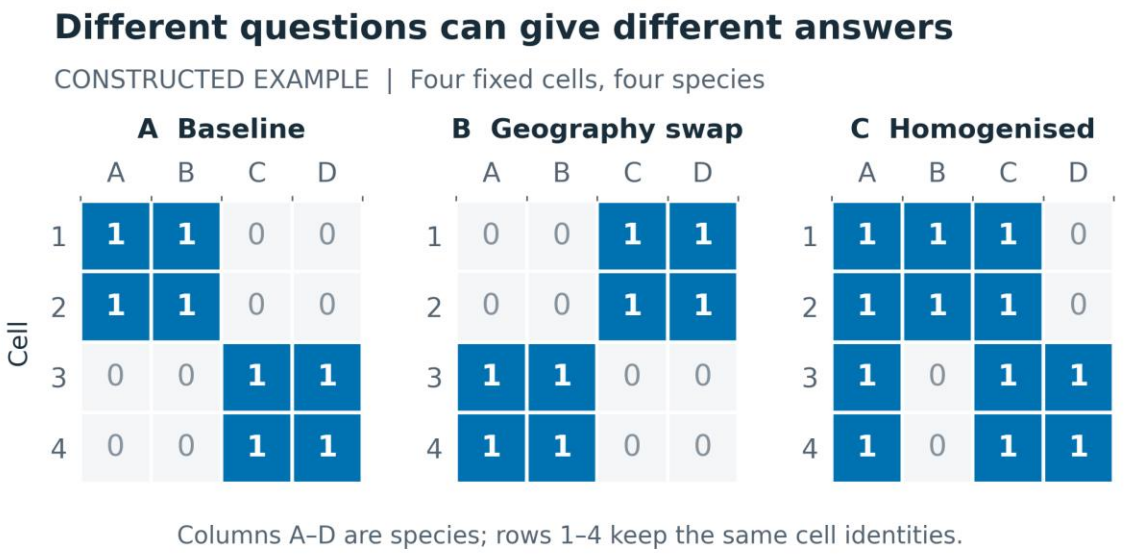
5. Plot persistent presences, losses and gains as stacked bars on identical scales. Show persistent absences separately if they would dominate the graphic. Supply a compact table with all four outcomes.
6. Inspect taxonomic and coverage sensitivities. Report the primary mask and species pool with every national percentage.
7. Write the biological statement before adding tests: “Records changed in this fraction of species–cell combinations, with this balance of gains and losses.”

## My thougths on the empirical precedent and interpretation

Read this paper (summary by Chatgbt given my direction for what is relevant for us): Terry, Langdon and Rossberg (2023) analysed repeated bird-atlas data for 80 species in 348 central English 10 km cells. Their wider study concerned codistribution, but a useful descriptive result was that 27% of occurrence entries changed between the earlier and later atlases, while total occupancy changed only from 46.5% to 46.0%. This is a close precedent for showing that nearly stable totals can conceal substantial rearrangement. It is not a precedent for our proposed national significance test, and we do not need to adopt its codistribution analysis.

In the reproducible example, 112 records are lost and 88 gained from Atlas 1 to Atlas 2. Thus 200 of 2,304 entries change, H = 8.68%, while the net loss is only 24 records. Whole-matrix Jaccard dissimilarity is 16.41%. A Results sentence could read: “Recorded occurrence changed in 8.68% of species–cell entries; losses and gains were broadly balanced.” The Discussion should say that this balance conceals redistribution, not that the assemblage was stable.

Whole-matrix Jaccard is also not the unweighted mean of cell-level Jaccard values. It weights each cell by the number of species in its temporal union. That is appropriate for the complete occurrence matrix, whereas an average across cells gives each cell equal importance.




| Exact metric | Baseline | Swap | Homogenised |
|---|---|---|---|
| Matrix Hamming distance (changed / all entries) | **0/16 = 0** | **16/16 = 1** | **4/16 = 1/4** |
| Mean temporal Jaccard (vs baseline) | **0** | **1** | **1/3** |
| Occupied cells per species (A, B, C, D) | **2, 2, 2, 2** | **2, 2, 2, 2** | **4, 2, 4, 2** |
| Species per cell (each of the four cells) | **2** | **2** | **3** |
| Spatial mean Jaccard (within each matrix) | **2/3** | **2/3** | **1/3** |

Temporal Jaccard compares each cell with itself through time.
Spatial mean Jaccard averages the six unordered pairs of cells.

Filled entries = presence (1); pale entries = absence (0).

Figure 1. Hypothetical matrices showing why the questions differ. Rows retain their geographical identity. The complete swap changes every entry while preserving each species' occupied-cell count, every cell's richness and mean spatial Jaccard dissimilarity. The third pattern has richer but more similar communities. These are constructed examples, not successive Danish atlases. Generated by Chatgbt by my instructions

# Question 2. Has the distribution of Danish range sizes changed?

## Biological question and measurement

Here we ask whether Denmark's breeding birds have shifted collectively towards more restricted or more widespread recorded distributions. Define range size as the number of comparable cells occupied by a species, and optionally divide by N.

$$r_{s,t} = \sum_{i=1}^{N} X_{i,s,t}\,, \qquad p_{s,t} = \frac{r_{s,t}}{N}.$$

$r_{st}$ is an occupied-cell count and $p_{st}$ is proportional occupancy for species s in atlas t. "Range-size frequency distribution" and "occupancy frequency distribution" describe the frequency of these values across species in this setting. I would use RSFD, while defining its operational meaning once.

This is Danish occupancy extent at 5 km resolution. It is not global range size, abundance, population size or the area of suitable habitat occupied. A species recorded in twenty coastal cells does not necessarily occupy 500 km² of habitat.

Use the fixed union of species and retain meaningful zeros. If a species disappears from the records, dropping it from the later histogram removes precisely the event we want to describe. A supplementary distribution conditional on being present in each atlas answers a different question: the sizes of ranges among that period's recorded breeders.

## Figures and effect sizes

Use three complementary figures. 1) "Identical-bin histograms" show how many species fall into occupancy classes. Define classes explicitly, including zeros, and check another bin width. 2) "Overlaid empirical cumulative distributions" show the fraction of species at or below every occupancy threshold without relying on bins. A curve lying lower over most thresholds generally indicates larger occupancies. Crossing curves indicate that different parts of the distribution moved differently. 3) "Species-matched scatterplots**"** put the same species on both axes; points above the equality line gained occupied cells.

You can derived many statistics from this. I suggest reporting the median, 10th and 90th percentiles, number of zeros, and proportions in a few stated classes, for example 0%, >0–10%, >10–50% and >50% occupancy. These thresholds illustrate reporting choices, not biological boundaries. Keep axes linear in the main figures. If a supplementary log scale helps the rare-species tail, show zero as a separate class or clearly label a log(1 + r) transformation.

For a distribution-wide effect size, use Wasserstein-1 distance:

$$W_1(t,u) = \frac{1}{S}\sum_{k=1}^{S} \left| p_{(k),u} - p_{(k),t} \right|.$$

The brackets indicate sorted values, not matched species identities. For equal-weight species, $W_1$ is the mean distance between corresponding ranks in the two distributions. Its units are occupied cells when using r, or proportion of the grid when using p. $W_1 = 0.03$ on the latter scale means a three-percentage-point average displacement under optimal matching of the distributions. It does not mean every named species changed by that amount.

Also calculate the mean signed occupancy change of named species and their mean absolute change, M. With a fixed grid and species pool:

$$0 \le W_1(t,u) \le M_{tu} = \frac{1}{S}\sum_{s=1}^{S}\left|p_{s,u} - p_{s,t}\right| \le H_{tu}.$$

The first inequality follows because sorting gives the least costly matching. The second follows because gains and losses can cancel in a species' net range change. This gives us a useful hierarchy: distributional change can hide species identity changes, and range-size changes can hide geographical movement.

Mean occupancy adds no independent information about total records or mean cell richness:

$$\bar{p}_t = \frac{R_t}{NS}, \qquad \bar{r}_t = \frac{R_t}{S}, \qquad \bar{\alpha}_t = \frac{R_t}{N} = S\bar{p}_t.$$

Here R is total recorded presences and $\bar{\alpha}$ is mean species richness per cell. The RSFD's shape, tails and zeros are therefore what it adds beyond those totals. For instance, a stable mean can coexist with increasing restriction of already narrow-ranging species and expansion of widespread species.

## What would a statistical comparison test?

Our primary RSFD analysis should be descriptive, with matched species changes alongside distributional distances. For the defined national species pool, these are complete finite-system summaries.

An omnibus equality test would ask whether the marginal occupancy distributions are identical across periods. The same species are measured repeatedly, so the ordinary independent-sample Kolmogorov–Smirnov test is not an automatic solution. A Friedman test addresses a repeated rank/location question under its assumptions; it does not test equality of all distributional shapes.

**Have a look this paper** (summary by Chatgbt given my direction for what is relevant for us):Martínez-Camblor, Carleos and Corral (2014) develop a repeated-measures Cramér–von Mises comparison of marginal distributions. Their approach retains within-subject dependence and uses a suitably centred bootstrap; the empirical application compares relative GDP per capita across 30 European countries in 2000 and 2011. This is a methodological starting point, not a verified ready-made test for our discrete, tied occupancy values and dependent species. Their random-sample framework requires adaptation before treating Denmark's complete bird list as independent sampled subjects.

If we pursue it, resampling whole species histories preserves each species' three measurements but breaks dependence among species. Resampling whole spatial blocks of all species and all atlases preserves more of the shared geographical dependence and changes the inferential target to spatial-process variation. Either choice needs justification. Bootstrapping $W_1$ around the observed difference supplies neither an automatic zero-centred null nor a calibrated equality test. I would require simulation checks before reporting a full-distribution p-value.

A systematic shift in mean occupancy is a simpler, different target and can be assessed with paired spatial-block uncertainty under the chosen spatial model. Practical similarity of RSFDs could instead mean $W_1$ below a pre-agreed tolerance; inferential support would require a calibrated upper confidence bound below it. A null fixing every species' range size fixes the RSFD exactly and cannot test its change.

## Cookbook (I asked chatgbt to generate on how to do the analyses I suggest)

1. Calculate column sums and column means from each matrix, retaining the same S species and all genuine zeros.
2. Produce the three histograms, ECDF overlay and all three matched scatterplots. Use the same scales.
3. Tabulate median, tail quantiles, zeros and explicit occupancy classes. Calculate $W_1$, mean signed change and M for every atlas pair.

4. Inspect named species with the largest changes. Check whether a change in the lower tail reflects contractions, disappearances, arrivals or combinations of these.
5. Repeat the distribution plots conditional on presence only as a labelled sensitivity analysis. Check histogram-bin sensitivity.
6. Decide whether the chapter needs description, a mean-shift interval, a practical-similarity assessment or an omnibus equality test. State the corresponding inferential target.
7. If testing distributions, develop and calibrate the repeated-data procedure before applying it. Do not substitute an independent-sample test simply because software offers one.
8. Report distributional and identity-sensitive results together. Base R's colSums(), quantile(), hist() and ecdf() cover the summaries; ggplot2 provides geom_histogram() and stat_ecdf(). transport::wasserstein1d(..., p = 1) calculates $W_1$, not a significance test.

## My reading of what previous papers actually establish on this topic

**Graves and Rahbek (2005)** – wonderfull paper 😊 - examined South American birds in 1° quadrats, using range-size distributions to investigate source-pool geometry and assemblage structure. They showed substantial variation in assemblage range-size properties. It provides spatial and conceptual motivation for our RSFD figure, but contains no repeated atlas comparison and no test of temporal equality. Thus, our analyses will be novel compared to this paper

**Suhonen and Jokimäki (2019)** compared 35 wintering bird species in 29 Finnish urban plots across three winters. They fitted alternative ranked occupancy-curve forms and compared support using AICc. The same broad symmetric bimodal form was preferred in each period despite changes in individual species' occupancies. This is a particularly relevant three-period bird example, but common fitted shape does not establish identical empirical distributions.

**O'Sullivan, Terry and Rossberg (2023)** examined annual occupancy distributions of river invertebrates, plants and diatoms over long series. They assessed trends in moments using Mann–Kendall tests and Sen slopes, alongside further modelling. Broad distributional properties were temporally robust despite turnover underneath them. This supports comparing aggregate structure with species identities. It tests selected moments, however, and its long time-series methods should not be transferred directly to three atlases.

## My interpretation of what all of this could mean

Suppose species A occupies 10 cells and B occupies 100. Later A occupies 100 and B occupies 10. The RSFD and $W_1$ are unchanged, but each named species changes by 90 cells. Alternatively, let each species retain its count while replacing its occupied cells with other cells. Then both the RSFD and matched range sizes are stable while the labelled occurrence pattern changes.

In our three decade time-series (assigning some theoretical examples of potential figures), Atlas 1→3 has $W_1 = 0.0842$ and $M = 0.0859$, with mean occupancy increasing by 0.0451. Say "The mean absolute distance between corresponding occupancy ranks was 8.42 percentage points, while named species changed by 8.59 points on average in absolute terms." Do not translate $W_1$ into an average population increase or geographical distance. As you remember from our first meeting, I think average figures is difficult to interpretative for data like these atlases with so many records and 200 species.

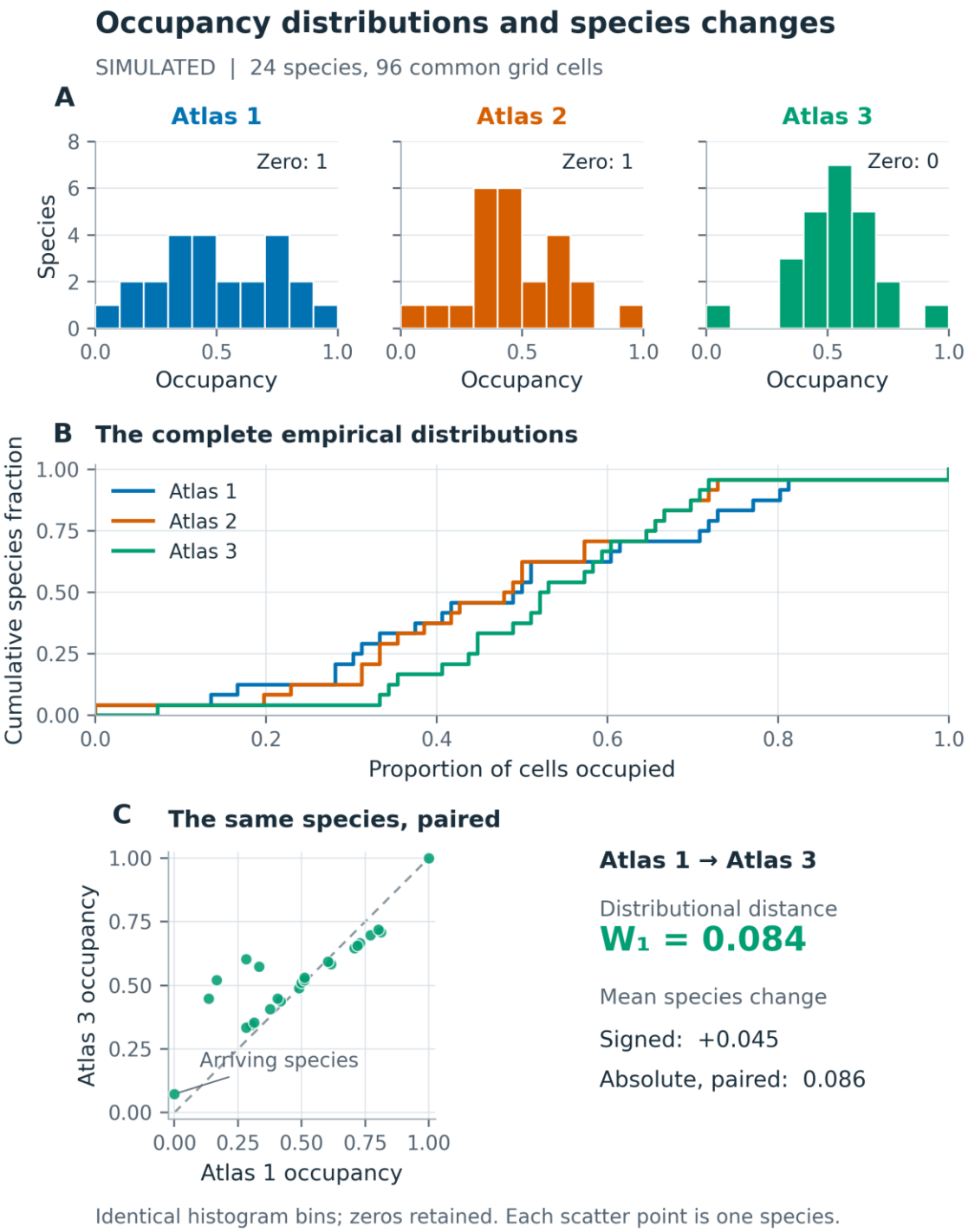



Figure 2. Simulated occupancy distributions and named-species changes for 24 species in 96 cells. Common histogram bins permit comparison; the ECDF avoids binning; the scatterplot retains species identities. All values come from the reproducible R example. These are not Danish results. (Illustrative figure generated by Chatgbt based on my input)

# Question3. Did the collective geographical occurrence pattern change systematically?

## Biological question and primary model

Question 1 counts every recorded mismatch. Here we could ask whether change has a collective temporal structure: national species occupancy frequencies changing, geographical patterns changing at specified scales, or both.

Retain the raw binary responses as the primary analysis. Squared Euclidean distance between two binary cell records equals the number of species mismatches. This gives equal weight to each presence/absence entry and matches our main question. Do not scale every species to unit variance without intending the resulting upweighting of some rare species.

Our proposed adaptation uses multivariate linear projection, equivalent to partial redundancy analysis: each cell has a species-specific baseline; atlas is a three-level factor; and a reduced set of geographical variables interacts with atlas. Space represents location and scale, not an explanatory environmental driver here.

| Model | Terms | What it allows |
|---|---|---|
| Cell only | Cell baselines | No modelled temporal component. |
| Average change | Cell + atlas | Species-specific national occupancy changes. |
| Full | Cell + atlas + space × atlas | National change plus geographically structured redistribution. |

The joint comparison is the full model against the cell-only model. The interaction comparison asks whether allowing geographically varying change improves on national average change alone. “National average” means the whole vector of species occupancy frequencies, not simply their sum. Some species can expand and others contract while their changes cancel in total richness. Okay, I here find myself suggesting the use of “average” 😊

Choose a modest set of spatial basis vectors from geography and the intended scale before examining temporal responses. These can be smooth coordinate terms or selected spatial eigenfunctions. Use projected coordinates measured in meaningful distances. A low-rank representation can miss fine-scale rearrangement; using every possible cell × atlas interaction would saturate an unreplicated design and leave no residual variation for a test.

## My suggestion for an understandable calculation for three periods (equations from chatgbt)

The appendix removes cell baselines with two orthogonal temporal contrasts:

$$D_1 = \frac{X_2 - X_1}{\sqrt{2}},$$
$$D_2 = \frac{2X_3 - X_1 - X_2}{\sqrt{6}}, \qquad D = [D_1, D_2].$$

Then it regresses the columns of D jointly on an intercept and centred spatial basis E. $D_1$ compares Atlas 2 with Atlas 1; $D_2$ compares Atlas 3 with the average of the first two. Together they retain all three-period temporal information. $D_2$ is not an acceleration estimate, particularly when intervals differ.

Partition the squared magnitude of D into national-average, geographical-interaction and residual sums of squares. Divide each component by total within-cell temporal variation. In the example, national-average change accounts for 7.8%, the selected spatial interaction for 13.7%, and their joint contribution for 21.5%. This denominator is temporal variation within cells, not all spatial and temporal variation in Danish bird communities.

## What supports an inferential claim?

The joint null says that all modelled temporal coefficients are zero. The interaction null allows national average changes but sets all geography × atlas coefficients to zero. Neither null says that every observed matrix entry is identical.

Spatial dependence remains after removing a cell intercept. Ordinary permutations of individual cells are therefore unsuitable when nearby residual histories remain correlated. Anderson and ter Braak (2003) show why choosing exchangeable units and the correct nuisance model is essential in multifactor permutation tests.

A block multiplier bootstrap, not a validated Danish-atlas test. It adds full-model residuals to a mean fitted under the relevant null, sharing one random multiplier across every species and both contrasts within a spatial block. Residuals receive a pointwise HC2 leverage adjustment. The statistic is extra fitted sum of squares, not an F statistic. This keeps the proposed within-block covariance together, while assuming enough approximately independent blocks.

Cluster wild-bootstrap theory provides background (see for example Djogbenou et al., 2019), but does not directly validate this multivariate spatial statistic. Before final inference, evaluate false-positive rates with simulated binary fields resembling the atlas design under explicit null models. Vary block size, layout, spatial range and species dependence. Sixteen demonstration blocks and a successful code run do not establish calibration.

In particular, jointly changing the signs of two contrasts is not a permutation of three atlas labels. It is not an exact test merely because the data have three periods. A literal atlas-label permutation would require exchangeability of the entire temporal history being permuted, a stronger assumption than unchanged means.

zzz

## Cookbook (I asked chatgbt to generate on how to do the analyses I suggest)

1. Use the aligned matrices and cell coordinates. Fit the joint three-atlas analysis before interpreting pairwise patterns.
2. Define the geographical scales of interest and construct a full-rank, centred spatial basis. Keep its dimension well below the number of independent spatial units.
3. Calculate $D_1$ and $D_2$ and fit the intercept-plus-space model. Partition temporal sums of squares. The appendix implements this with base R QR projections.

4. Inspect maps of residual contrasts and their spatial dependence. Check whether a few species or regions dominate the result.
5. Specify the exact null for each comparison, including allowed average change for the interaction test.
6. Develop the block dependence model and calibrate the proposed bootstrap. Once adequate, increase simulation precision and report the procedure and diagnostics. Until then, use the fitted fractions as descriptive effect sizes.
7. Calculate analogous pairwise partitions for 1→2, 2→3 and 1→3. If conducting three additional tests, define the family of comparisons and adjust accordingly.
8. Plot the partition of temporal variation and maps of fitted changes. If using ordination, place all atlases in one common coordinate system. Independently rotated ordinations do not give comparable trajectories.
9. As an optional sensitivity, use Hellinger-transformed responses and explain the changed question. A presence becomes $1/\sqrt{}$(cell richness), changing the weight of species records in species-rich cells (Legendre and Gallagher, 2001).

vegan::rda() supports multivariate constrained and partial ordination. adespatial::stimodels(model = "5") implements the published space–time approach, but its default temporal eigenvectors need not span both contrasts for three periods. Supply full temporal coding. Neither its default permutation procedure nor permute's restricted-permutation constructors automatically supplies a justified spatial test. The explicit contrast calculation is easier to audit here.

## My understanding of precedent and biological interpretation

**Legendre, De Cáceres and Borcard (2010)** – again my gavorite ecological statisticians 😊 - asked how to test changing spatial community structure when sites are revisited without replication within each site and date. Their model 5 retains full site and time effects with a reduced interaction. In a forest application with 315 tree species in 1,250 fixed 20 × 20 m cells and four censuses, spatial eigenfunctions interacted with full temporal contrasts. For Hellinger-transformed abundances, the community interaction was significant ($p = 0.001$; 999 permutations), indicating changing geographical composition. This is a strong methodological and empirical design precedent, although it uses forest abundances rather than bird-atlas occurrences. Its approximate permutation procedure does not establish validity for arbitrary spatial dependence.

Within the fitted component, a large average term and a small interaction indicate that changes in national occupancy frequencies dominate over geographically structured changes at the represented scales. A substantial fitted interaction describes recorded changes that vary by location; a calibrated test can assess a systematic component under its null. A small fitted joint fraction alongside large H suggests that the chosen model captures little of extensive recorded change, perhaps because it is local or irregular. It does not establish stability.

A suitable Results sentence is: "The selected spatial interaction described 13.7% of within-cell temporal variation in the simulated records." The biological interpretation is geographical restructuring at those modelled scales. Fitted binary-response contrasts are not probabilities, population trends or evidence of interactions among species.

# Question 4. Which grid cells changed unusually strongly?

## Measure local temporal change first

We can compare the community in one cell with that same cell in a later atlas. Let $a_i$ be species retained, $b_i$ species lost and $c_i$ species gained in cell i.

$$TBI_i = \frac{b_i + c_i}{a_i + b_i + c_i}, \quad L_i = \frac{b_i}{a_i + b_i + c_i}, \quad G_i = \frac{c_i}{a_i + b_i + c_i}.$$

$TBI_i$ is temporal Jaccard dissimilarity. $L_i$ and $G_i$ are its loss and gain contributions and add to $TBI_i$. Also retain raw counts, because a change of two species in a small assemblage can produce a large proportion.

For example, a cell with 15 species initially and 15 later can retain ten, lose five and gain five. Its richness is unchanged, but TBI = 10/20 = 0.5. Half the species in the temporal union occur in only one period.

Loss/gain components preserve temporal direction. Replacement/nestedness components instead partition kinds of compositional difference. They are not interchangeable. If needed, add the Baselga (2010) partition as a supplementary analysis, explaining that nestedness-resultant dissimilarity is a metric component, not proof of an extinction mechanism.

## We need to define "unusual" before testing it

I recommend a baseline-fixed conditional random-change null as our primary candidate. For each atlas pair and each species, retain the observed earlier map and the exact numbers of recorded losses and gains. Randomly allocate losses among its initially occupied cells and gains among its initially unoccupied cells.

This asks: given how much each species changed nationally, was change unusually concentrated in this cell? It does not ask whether national change occurred at all.

The null preserves both endpoint range sizes and each species' national transition totals. It does not preserve later cell richness, later geographical clustering or dependence among species' changes. Those are explicit choices, not innocuous technical details. If unrestricted allocation across Denmark is an inappropriate baseline, repeat within pre-defined geographical strata, preserving transition counts within them. This changes the null to concentration beyond those regional constraints.

The three pairwise nulls each condition on their own earlier atlas. They do not jointly simulate a coherent three-period history. Use them for the stated separate comparisons; a joint temporal null would require additional construction.

## Cookbook and software (generated by Chatgbt)

1. Calculate TBI, loss/gain counts and components for every cell in each pair. Investigate missing or double-empty temporal comparisons before continuing.
2. Map the observed values with the same colour scale across pairs. Show losses and gains separately and label net richness change as a different quantity.
3. Document the null's fixed quantities and exchangeable locations. Decide whether a national or stratified allocation is primary.
4. Generate later matrices by the conditional allocation rule. Recalculate TBI in every replicate. Check that each species' losses, gains and range sizes are preserved exactly.

5. Report each cell's observed TBI, mean null TBI and excess over that expectation. For an upper-tail test, use (1 + number of simulated values at least as large)/(B + 1), where B is the number of simulations.
6. Define a multiple-testing family. The appendix applies Holm adjustment across all cells in all three comparisons. Holm can accommodate dependent tests if the individual p-values are valid; it cannot repair a biologically inappropriate null.
7. Set simulation precision accordingly. For a family of 6,000 cell-by-contrast tests, the first Holm threshold at 5% is about 0.0000083. With 9,999 simulations, the smallest plus-one p-value is 0.0001, so no first rejection is possible. At least 119,999 simulations are needed merely to reach that first threshold; more are needed for useful precision. Use enough simulations, a justified alternative multiplicity strategy, or continuous departures without binary significance claims.
8. Check sensitivity to geographical restrictions and the empty-community policy. Do not silently discard inconvenient null replicates. Report whether "exceptional" means large observed change, large excess over the null, or multiplicity-adjusted evidence.

The custom random_later() function implements our baseline-fixed null. adespatial::TBI() calculates temporal dissimilarity and has its own published randomisation, which answers a different conditional question. Its documented interface does not provide our spatially stratified allocation automatically.

### Methodological and empirical precedents as I have found it

**Legendre (2019)** develops TBI, loss/gain components and tests for exceptionally changed sites, illustrated with repeated ecological surveys. Its published randomisation independently permutes each species among sites, using the same permutation at both times. This preserves that species' paired temporal histories but relocates the baseline as well. It identifies exceptional sites relative to the overall temporal pattern. Our proposed null instead keeps the earlier geographical map fixed.

**Baselga, Bonthoux and Balent (2015)** compared 47 bird species at 252 French agricultural sites in 1982 and 2007. They partitioned temporal dissimilarity and compared observed change with frequency-constrained randomisations. Their findings illustrate substantial local temporal change alongside little change in regional heterogeneity. The study is a close empirical precedent, but its randomisation did not fix each species' observed loss and gain totals as ours does.

Imagine two cells both have TBI = 0.5. Null means of 0.45 and 0.10 imply excesses of 0.05 and 0.40. The second has the larger excess, but statistical exceptionality depends on each cell's full null distribution, including its spread and tails. Report departures from the specified allocation model without assigning an environmental cause.

## Question 5. Did Danish assemblages become more homogeneous or heterogeneous over time?

### Biological question and primary metric

This comparison is spatial within each atlas, followed by a temporal comparison of those spatial differences. Are two Danish cells, considered under a fixed weighting and scale, more similar in their species lists in later atlases?

Calculate Jaccard dissimilarity between every unordered pair of different cells and average it:

$$\bar{\beta}_t = \frac{2}{N(N-1)} \sum_{i<j} d_J\left(X_{i,\cdot,t}, X_{j,\cdot,t}\right),$$
$$\Delta\bar{\beta}_{tu} = \bar{\beta}_u - \bar{\beta}_t.$$

$\bar{\beta}_t$ is mean spatial Jaccard dissimilarity. A negative Δβ indicates observed homogenisation; a positive value indicates heterogenisation. This is the primary national metric I recommend because its meaning is direct and it retains information about how species are distributed among cells.

Do not treat the N(N − 1)/2 distances as independent replicates. Many share the same cells, and neighbouring cells are correlated. Nor should a temporal TBI value be called spatial beta diversity: one compares a cell with itself over time; the other compares different cells within one time.

The hypothetical complete swap in Figure 1 gives TBI = 1 in every cell while mean spatial dissimilarity remains 2/3. The local communities change completely, but their degree of spatial differentiation remains the same. In the third pattern, mean richness rises from two to three species and mean spatial Jaccard declines to 1/3. Homogenisation therefore need not involve declining local richness.

## Uncertainty and random expectation answer different questions

First calculate the observed change for all three pairs. If uncertainty is useful, apply a paired spatial-block bootstrap: resample whole blocks, carrying every species and all three atlas records together, and recalculate $\bar{\beta}$ and its differences. This retains temporal pairing and the within-block spatial structure in each selected history.

The resulting intervals require a spatial-process interpretation and sufficiently representative, weakly dependent blocks. A fixed national census has no design-based sampling error simply because a bootstrap can be run. Choose block sizes using residual or change-pattern dependence, examine several reasonable layouts, and allow for regional nonstationarity, for example through stratified resampling. A bootstrap cannot correct unknown differences in detection.

Second compare the observed Δβ with the conditional random-change null from question 4. For each simulated later matrix, recompute spatial beta diversity and its difference from the fixed baseline. This asks whether the geographical allocation of observed changes produced more or less spatial differentiation than that allocation model predicts.

Keep the sign and the null comparison separate. An observed decline may be entirely ordinary under the null. A positive change can nevertheless lie unusually far below a null that predicts a still larger increase. That is “less heterogenisation than expected”, not observed homogenisation. Report observed Δβ, the null distribution and the relevant directional comparison together.

## This is important!: richness, range sizes and spatial scale

Increasingly widespread species can make cells share more species. That creates a useful connection to RSFD change, but does not establish a causal explanation. The fixed-transition null preserves species’ endpoint range sizes, and therefore the endpoint RSFD and mean cell richness. Any remaining variation in mean pairwise Jaccard comes from how occurrences are arranged.

Some national beta measures cannot answer that conditional question. Whittaker’s $\gamma/\bar{\alpha}$ is fixed when every species’ range size is fixed: national richness γ and total occupancy are then fixed. It is a useful descriptive index for some purposes, but supplies no varying statistic for this particular allocation null.

If we additionally fix each cell’s later richness, we ask whether arrangement matters beyond those local totals. That is an optional, more constrained question requiring a separately validated sampler. Chase et al. (2011) provide methodological guidance on richness-conditioned dissimilarity, showing why raw

compositional difference and deviation from a richness-based expectation can tell different stories. Their procedure is not our temporal transition null.

National all-pairs averages mix nearby and distant comparisons. A complementary local analysis can average dissimilarity within fixed neighbourhoods or fixed distance bands. Hold neighbourhood definitions, cell membership and edge rules constant across atlases. Overlapping neighbourhoods remain dependent. A national average can conceal local homogenisation in some regions and heterogenisation elsewhere.

## Cookbook and software (generated by Chatgbt)

1. Compute each atlas's mean pairwise spatial Jaccard on the common grid. Check empty communities in observed, resampled and simulated matrices under one explicit rule.
2. Calculate the three signed differences. Plot the atlas values and a separate difference plot with any justified uncertainty intervals.
3. If the matrix is large, calculate pairwise distances in chunks. Alternatively, define one reproducible sample of cell pairs and use it consistently, quantifying the additional approximation error.
4. For spatial-process intervals, resample the same complete cell histories in all periods using spatial blocks. Recompute the full statistic; do not bootstrap the list of pairwise distances.
5. Run the fixed-transition null separately for each atlas pair. Plot its $\Delta\beta$ distribution with the observed value. Decide directional hypotheses in advance, or use an appropriate two-sided comparison with multiplicity adjustment.
6. Optionally partition spatial dissimilarity into replacement and nestedness-resultant components. betapart::beta.pair(..., index.family = "jaccard") returns total, replacement and nestedness components as beta.jac, beta.jtu and beta.jne. Apply the same pairing and uncertainty logic to their summaries.
7. Add a neighbourhood or distance-band analysis only where it helps interpret the national result. Explain the spatial scale in the figure caption.
8. Report the observed direction separately from departure from the null, and distinguish raw change from any richness-conditioned result.

vegan::vegdist(..., method = "jaccard", binary = TRUE) calculates spatial dissimilarities. The appendix's beta_mean() performs the same primary calculation in base R. Anderson et al. (2006) develop multivariate dispersion as an alternative beta-diversity representation; vegan::betadisper() implements that approach. It is useful as a sensitivity analysis, but is not identical to mean pairwise Jaccard. A multivariate group-location test also does not automatically establish a change in dispersion.

## Papers and interpretation as I have found and read them

**Baselga (2010)** develops additive separation of compositional dissimilarity into replacement and nestedness-resultant components, illustrating the distinction with European longhorn beetles. Its contribution is methodological and spatial, not a temporal bird-atlas test.

**Anderson, Ellingsen and McArdle (2006)** formulate beta diversity through multivariate dispersion and illustrate why within-group spread must be distinguished from shifts in assemblage location. This offers a methodological alternative and helps prevent interpreting every multivariate atlas effect as homogenisation.

**Wayman et al. (2024)** compared 159 bird species in 2,291 British 10 km atlas cells between 1988–1991 and 2008–2011. They used spatial neighbourhoods and beta-diversity partitions to separate contributions

of colonisations and extirpations. Homogenisation was widespread, with colonisations and extirpations often contributing in opposite directions. This is the closest national bird-atlas precedent here. Its neighbourhood measures and one-sample change tests are not our national mean-Jaccard statistic or our fixed-transition null.

Figure 3 below reproduces Baselga et al.'s original Figure 2, illustrating this distinction.

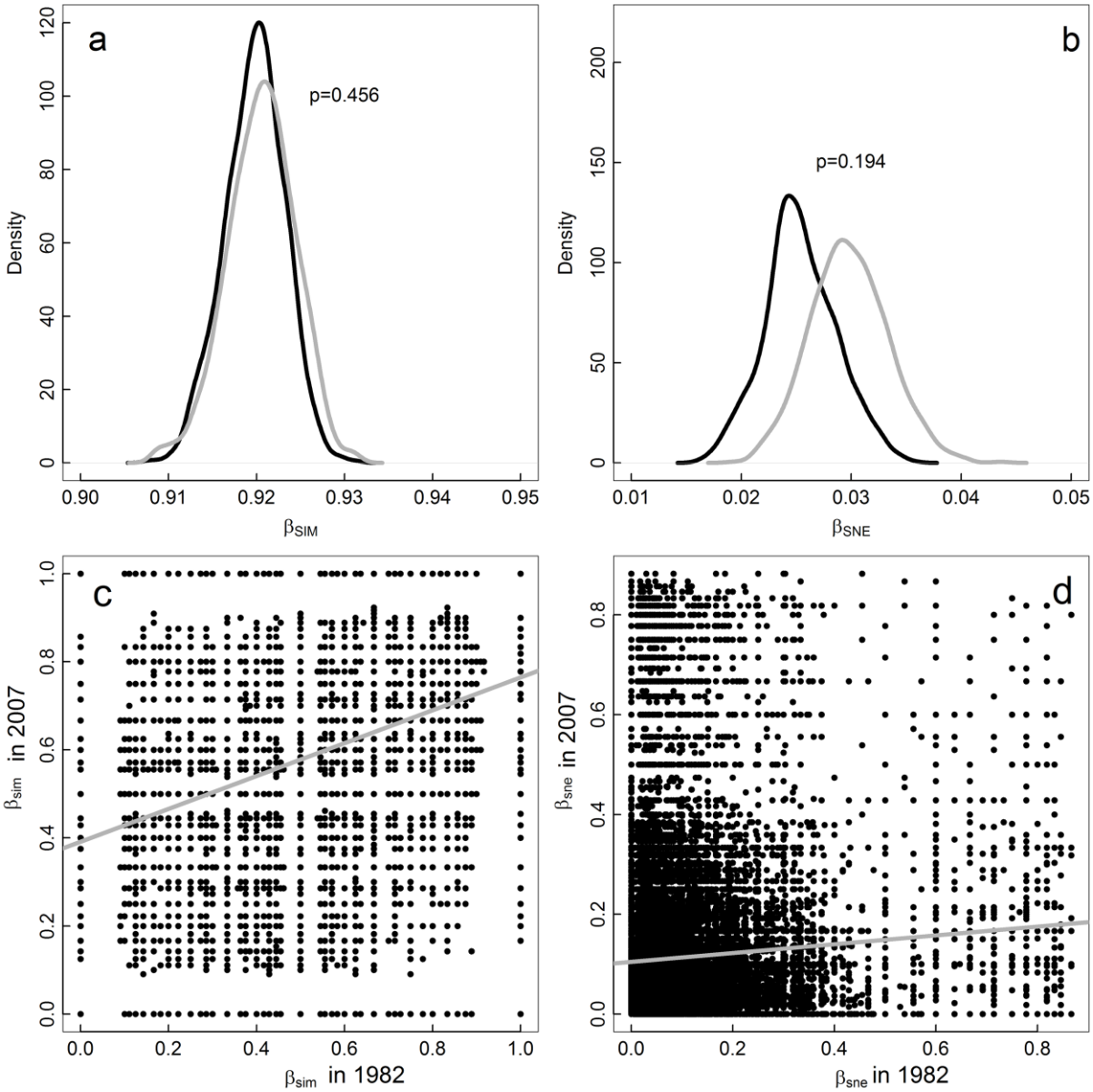



Figure 3. Published example of regional summaries concealing changes among particular sites. Upper panels compare turnover and nestedness-resultant spatial beta diversity in resampled sets of sites; lower panels compare the same site-pair dissimilarities between years. Reproduced unchanged from Baselga, Bonthoux and Balent (2015), original Figure 2, PLOS ONE 10:e0127913, DOI: 10.1371/journal.pone.0127913, under Creative Commons Attribution 4.0 (https://creativecommons.org/licenses/by/4.0/). These are French bird data, not Danish results. The published metrics differ from our primary mean Jaccard; non-significant differences do not prove equivalence. Figure reproduced by Chatgbt

In the simulated example, mean spatial Jaccard is 0.562, 0.602 and 0.583 in the three atlases. Thus the first interval shows heterogenisation, the second homogenisation, and the full period a net increase of 0.020. Meanwhile, 19.53% of complete matrix entries change from Atlas 1 to Atlas 3. Both statements can be true.

The illustrative paired-block interval for the full-period difference is approximately −0.012 to 0.051. Under the example's spatial-process assumptions, this interval spans both signs; it does not establish equivalence. The positive observed change is not unusually large relative to the conditional null (upper-tail Monte Carlo p = 0.110). These statements describe this simulation only, and the small number of teaching replicates is not a recommendation for final precision.

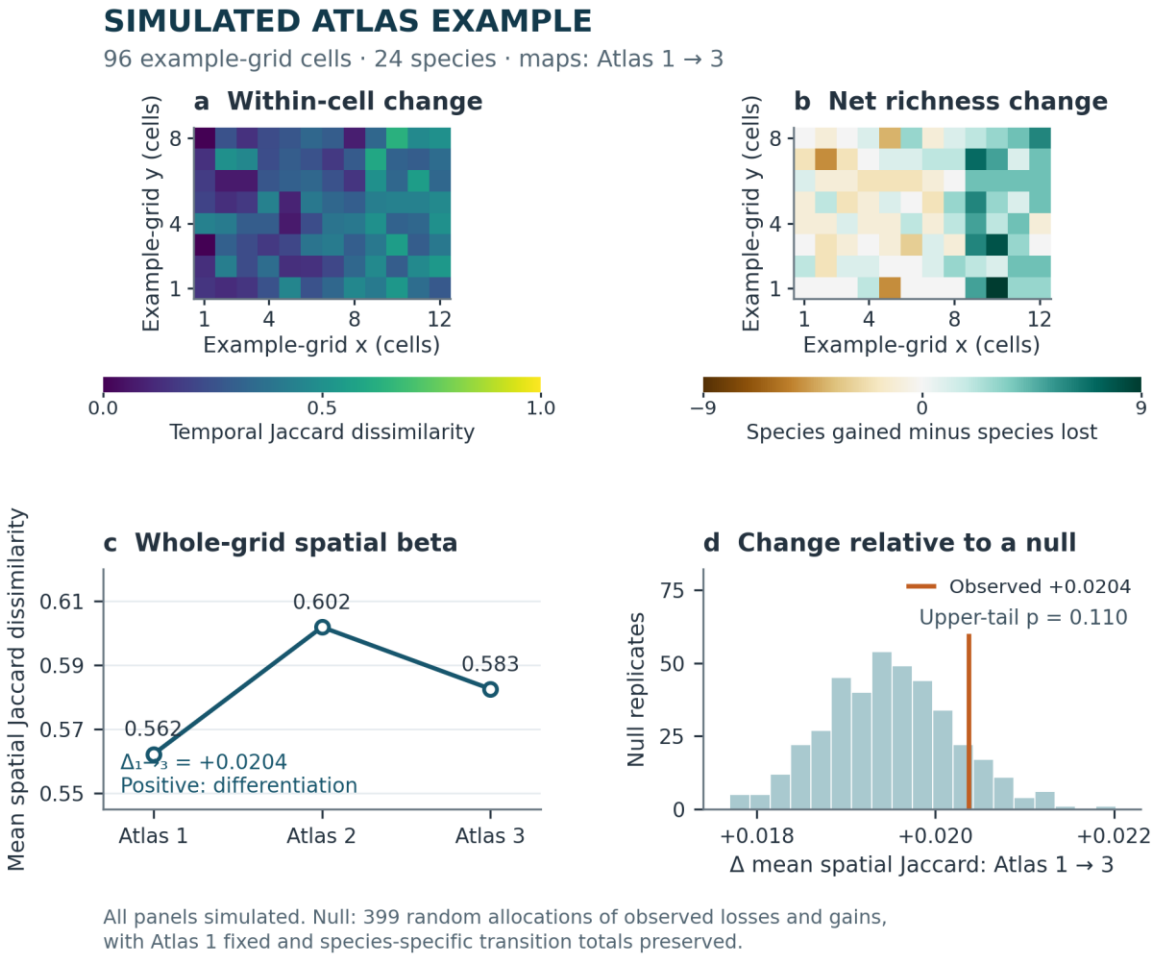



Figure 4. Simulated local and spatial change. The maps compare the same cells through time; the lower panels summarise differences among cells within each atlas and compare the full-period change with the baseline-fixed allocation null. The maps are an artificial rectangular grid. The observed sign of change and its position relative to the null answer separate questions.

# How I think the five answers can be brougthtogether

(we can also pick some of the give proposed question and do)

We could aim for a chapter that explains how a reader can move from a national summary to the species and places underneath it. The first task is to identify what changed, then establish the scale and conditional meaning of that change.

Many changed records with little net occupancy change indicate balanced gains and losses. Similar RSFDs with large matched range changes indicate redistribution of range sizes among species. Stable named-species range sizes with substantial matrix change indicate redistribution among cells. Large local TBI with stable spatial beta indicates changing communities without a comparable change in their overall differentiation. Homogenisation with stable or increasing richness indicates that cells are sharing more similar species lists, not necessarily becoming poorer.

None of these combinations identifies a climate, land-use or interaction mechanism. You have already done these analyses – and they wpuld come in the chapter after this. Each specifies a pattern that the subsequent explanatory analyses can address. At this stage, the strength of the chapter will come from making that pattern and its uncertainty clear.

| Question | Primary analysis | Main figure | Report | Biological conclusion | Principal limit |
|---|---|---|---|---|---|
| 1. Complete records | Direct transition counts | Persistence / gain / loss bars | H, J, gains, losses | Magnitude and balance of recorded rearrangement. | Does not separate detection change. |
| 2. Range sizes | RSFD plus matched species | Histograms, ECDF, scatter | $W_1$, quantiles, matched changes | Change in occupancy distribution and species identities. | Range size omits geographical location. |
| 3. Collective geography | Multivariate temporal partition | Fitted maps; variance partition | Average and spatial fractions; calibrated test | Change represented at selected spatial scales. | A reduced spatial model can miss local change. |
| 4. Local change | Temporal Jaccard + allocation null | TBI, gains/losses, excess maps | Observed TBI and null departure | Concentration of change beyond stated constraints. | Exceptionality is conditional on the null. |
| 5. Spatial similarity | Paired spatial beta + allocation null | Atlas beta; difference and null plots | Signed $\Delta\beta$, interval, null comparison | Species lists more or less similar on average. | Scale, richness and the null affect interpretation. |

## A suggestion to a manageable order of work

**First, finish the common data preparation and descriptive core:** question 1, the RSFD and matched-species summaries in question 2, observed local change in question 4, and observed spatial change in question 5. These calculations should provide the chapter's main figures and are immediately implementable once the matrices are ready.

**Second, add the conditional allocation analyses for questions 4 and 5**, after deciding what random placement means biologically. Check the spatial restrictions, empty-community policy and simulation precision. Their results should explicitly describe departures from that null.

**Third, complete the joint space–time partition in question 3 and develop its inference carefully.** The descriptive partition is straightforward. A calibrated test needs the real grid, an appropriate spatial basis and a justified dependence model. It is better to report an informative fitted effect size than to attach an unjustified p-value.

**Keep optional extensions selective:** Hellinger sensitivity, replacement/nestedness partitions, local beta diversity and a formal RSFD equality test should earn their place by answering a question the primary analyses leave unresolved.

When you write each Results paragraph, start with the quantity, its direction and its magnitude. When you write the corresponding Discussion paragraph, explain which birds or cells contributed and what level of biological organisation the result describes. That will keep the methods connected to the question that motivated them. Do this very briefly and centered around an illustrative items (a table, figure or the statistical results)

Best wishes,

Carsten

# Appendix A. Software and implementation (generated by chatgbt)

The principal workflow below is self-contained in base R. It generates 24 simulated species in 96 artificial grid cells, calculates every pairwise comparison, runs teaching versions of the resampling procedures, and writes results and basic plots to an atlas_outputs folder in the working directory. Replace only the simulated input section after completing the common-grid checks; retain species and cell identities.

The full script was executed successfully in R 4.6.0 through webR 0.6.0. Independent numerical checks also reproduced the multivariate sums of squares using the full stacked cell-and-atlas model. These checks establish calculation and code functionality. They do not establish the false-positive rate or interval coverage of the proposed inferential procedures for Danish atlases.

The package calls at the end are optional equivalents checked against current official documentation on 19 October 2026; they were not executed in this environment. A repeated-measures RSFD equality test is not implemented because its application needs further statistical adaptation. The block multiplier procedure is explicitly illustrative and must be calibrated before confirmatory use.

| Task | Functions or tools | Implementation status |
|---|---|---|
| Complete-matrix change | matrix_change(); base R sums | Executed. Direct counts and metrics. |
| Range-size comparison | colSums(), colMeans(), quantile(), ecdf(); W1() | Executed. Distribution distances and matched changes; no automatic equality test. |
| Space–time partition | contrast_data(), project(), wild_test() | Partition executed and checked. Bootstrap executes but remains an uncalibrated illustration. |
| Local change and null | local_change(), random_later(), p.adjust() | Executed. Conditional allocation and Holm adjustment; null assumptions remain substantive. |
| Spatial beta and intervals | jaccard_matrix(), beta_mean(); block resampling | Executed. Paired spatial-process intervals and separate conditional-null comparisons. |

## Implementation choices to retain when adapting the script

The teaching runs use 399 resamples or simulations for speed. The smallest plus-one p-value is therefore 0.0025, and the pooled local Holm tests are necessarily coarse. Increase precision for a real analysis; more replicates do not repair incorrect exchangeability or an inappropriate null.

The simulated grid has 16 equal blocks of six cells, and one species is present everywhere. Those conveniences are not recommended assumptions for Denmark. The spatial functions stop if a double-empty comparison is undefined. Decide an explicit empty-community convention for the real analysis, including null and bootstrap replicates. If two genuine empty lists are assigned Jaccard distance zero, document that absence of all breeders is being treated as compositional similarity and assess its effect.

The custom projection checks full column rank. Each reduced model must also be nested in the full model. The multiplier bootstrap uses full-model residuals, a pointwise HC2 adjustment and an

unstudentised extra-sum-of-squares statistic. It is not a conventional restricted-residual bootstrap, an exact temporal permutation or an automatic cluster-robust F test.

Each pairwise allocation null is separate. Do not combine its replicates into a supposed coherent three-period simulation or joint maximum-statistic test. Holm adjustment of the separate, valid cell-by-contrast p-values is a different operation and is implemented.

## Official software references

R Core Team: R statistical computing environment and manuals

vegan: binary dissimilarities, vegdist()

vegan: partial and constrained ordination, rda()

vegan: multivariate dispersion, betadisper()

adespatial: temporal beta diversity, TBI()

adespatial: space–time models and spatial eigenfunctions

betapart: beta.pair() and beta-diversity partitions

permute: how() and shuffleSet() construct restricted permutations

ggplot2: geom_histogram()

ggplot2: stat_ecdf()

transport: wasserstein1d() and optimal-transport distances

Restricted-permutation software generates the restrictions requested by the analyst; it does not establish that observations are exchangeable. Similarly, distance and ordination functions calculate their stated quantities, but do not automatically supply a temporal, spatial or detection-error model.

## Supplementary illustration of the executed calculations

The bars below link question 1's entry accounting with question 3's decomposition. Large numbers of changed records can coexist with a relatively modest fraction captured by the chosen geographical model. The R appendix produces basic versions of the principal plots; the document figures use the same saved numerical outputs with publication-style formatting.

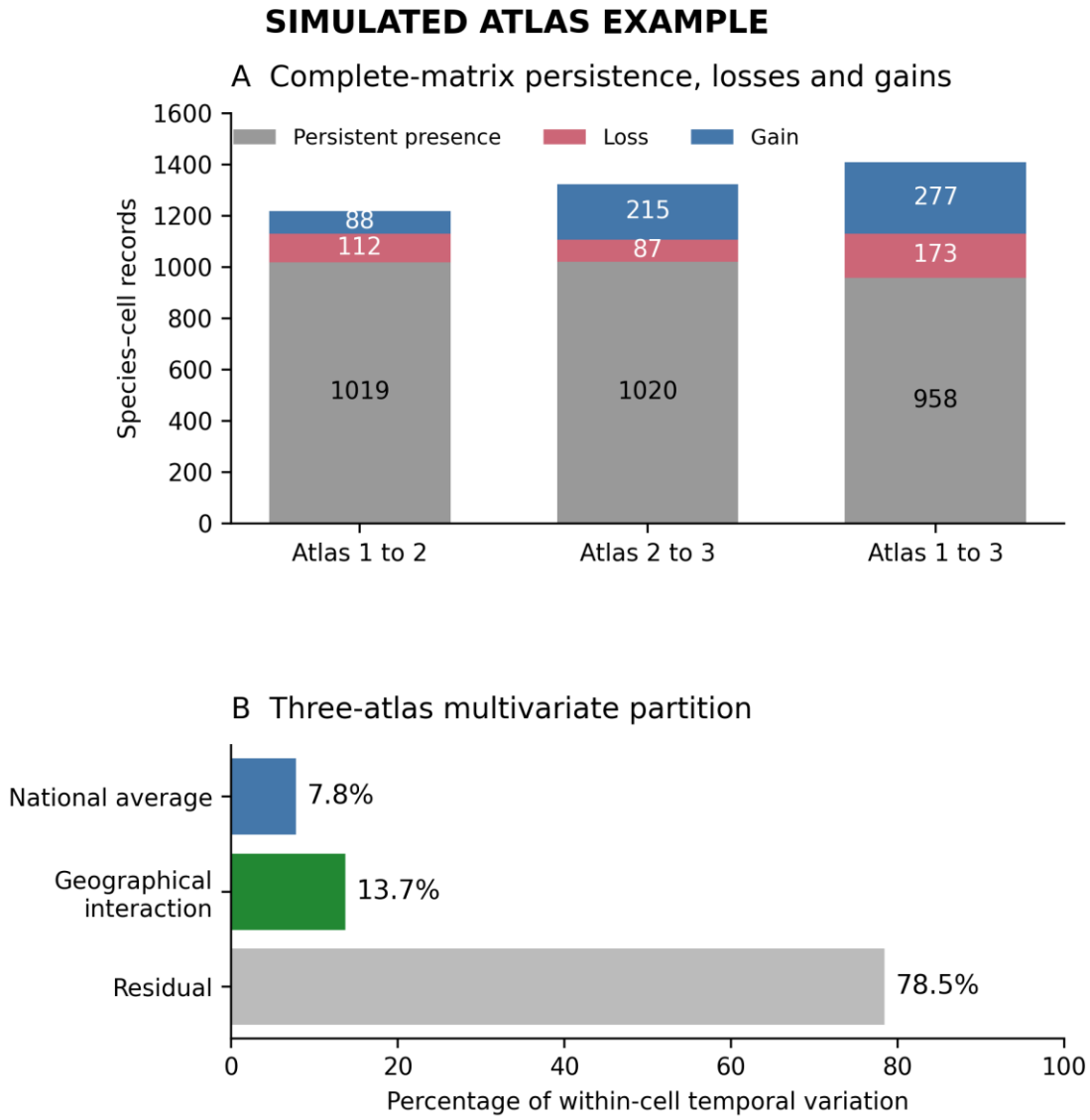



Figure 5. Simulated complete-matrix accounting and the three-period temporal partition. Persistent absences are omitted from panel A; full counts remain in the R output. Panel B partitions variation in the temporal contrasts, using five spatial coordinate terms. The fitted fractions are descriptive and do not validate the illustrative bootstrap.

# Appendix B. Reproducible R code

Copy the complete code below into a plain-text R script and run it in a working directory where output files may be created. No packages are required for the executed workflow. All example data, cell coordinates, species names and maps are simulated.

```
# Danish bird atlas teaching example. All data below are simulated.
# Base R implementation; see the letter for inferential assumptions.
set.seed(19012026)
dir.create("atlas_outputs", showWarnings = FALSE)
xy <- expand.grid(x = 1:12, y = 1:8)
N <- nrow(xy); S <- 24L
sp <- sprintf("Species_%02d", seq_len(S))
cells <- sprintf("Cell_%03d", seq_len(N))
base_p <- seq(0.08, 0.85, length.out = S)
eta <- outer(as.numeric(scale(xy$x)), rep(c(-0.65, 0.65), S/2))
P <- plogis(sweep(eta, 2, qlogis(base_p), "+"))
A1 <- matrix(rbinom(N*S, 1, as.vector(P)), N, S)
dimnames(A1) <- list(cells, sp)
A1[, 1] <- 1L                # Avoid empty cells in this example.
A1[, S] <- 0L                # A species arriving in Atlas 3.
pick <- function(v, k) v[sample.int(length(v), k)]
transition <- function(X, loss_rate, gain_rate) {
  Y <- X
  for (s in 2:ncol(X)) {
    on <- which(X[, s] == 1); off <- which(X[, s] == 0)
    L <- floor(length(on)*loss_rate[s])
    G <- floor(length(off)*gain_rate[s])
    if (L > 0) Y[pick(on, L), s] <- 0L
    if (G > 0) Y[pick(off, G), s] <- 1L
  }
  Y
}
A2 <- transition(A1, rep(0.12, S), rep(0.09, S))
A2[, S] <- 0L
A3 <- transition(A2, rep(0.10, S), seq(0.02, 0.24, length.out=S))
A3[xy$x >= 9, 2:5] <- 1L        # A constructed geographical shift.
A3[, S] <- 0L
A3[pick(seq_len(N), 7), S] <- 1L
atlases <- list(Atlas1=A1, Atlas2=A2, Atlas3=A3)
stopifnot(all(vapply(atlases, function(X) {
  identical(dimnames(X), dimnames(A1)) &&
    all(X %in% c(0,1)) && all(rowSums(X) > 0)
}, logical(1))))
pairs <- list(c(1,2), c(2,3), c(1,3))
pair_names <- c("1_to_2", "2_to_3", "1_to_3")
block <- interaction((xy$x-1)%/%3, (xy$y-1)%/%2, drop=TRUE)

# 1. Complete labelled occurrence matrices.
matrix_change <- function(X, Y) {
  a <- sum(X == 1 & Y == 1); b <- sum(X == 1 & Y == 0)
  c <- sum(X == 0 & Y == 1); d <- sum(X == 0 & Y == 0)
  c(persist=a, loss=b, gain=c, absent=d,
    changed=(b+c)/length(X), jaccard=(b+c)/(a+b+c),
    net_records=c-b)
}
changes <- t(vapply(pairs, function(k) {
  matrix_change(atlases[[k[1]]], atlases[[k[2]]])
}, numeric(7)))
rownames(changes) <- pair_names
print(round(changes, 4))

# 2. Range sizes and identity-sensitive comparisons.
```

```
occupancy <- sapply(atlases, colMeans)
W1 <- function(x, y) mean(abs(sort(x)-sort(y)))
range_change <- t(vapply(pairs, function(k) {
  x <- occupancy[,k[1]]; y <- occupancy[,k[2]]
  c(W1=W1(x,y), mean_signed=mean(y-x),
    mean_absolute_matched=mean(abs(y-x)))
}, numeric(3)))
rownames(range_change) <- pair_names
print(round(range_change, 4))
stopifnot(all(range_change[,"W1"] <=
  range_change[,"mean_absolute_matched"] + 1e-12))
stopifnot(all(range_change[,"mean_absolute_matched"] <=
  changes[,"changed"] + 1e-12))
range_summary <- t(apply(occupancy, 2, function(p) {
  c(zero=sum(p==0), mean=mean(p),
    quantile(p, c(.1,.5,.9), names=FALSE))
}))
colnames(range_summary) <- c("zero","mean","q10","median","q90")

# 3. Two orthonormal atlas contrasts remove each cell's intercept.
# Raw binary responses preserve the original occurrence weighting.
contrast_data <- function(xs) {
  cbind((xs[[2]]-xs[[1]])/sqrt(2),
        (2*xs[[3]]-xs[[1]]-xs[[2]])/sqrt(6))
}
z <- scale(as.matrix(xy))
E <- cbind(z[,1], z[,2], z[,1]^2, z[,2]^2, z[,1]*z[,2])
E <- scale(E, center=TRUE, scale=FALSE)
H <- cbind(Intercept=1, E)
D <- contrast_data(atlases)
project <- function(H, D) {
  if (ncol(H)==0) return(matrix(0,nrow(D),ncol(D)))
  q <- qr(H)
  if (q$rank < ncol(H)) stop("Design is not full column rank.")
  Q <- qr.Q(q); Q %*% crossprod(Q,D)
}
full_fit <- project(H,D); mean_fit <- project(H[,1,drop=FALSE],D)
SS <- c(total=sum(D^2), atlas=sum(mean_fit^2),
        interaction=sum((full_fit-mean_fit)^2),
        residual=sum((D-full_fit)^2))
print(round(SS, 4))
print(round(SS[c("atlas","interaction")]/SS["total"], 4))

# Optional approximate null-imposed block wild bootstrap, using an
# unstudentised extra-SS statistic and full-model HC2 residuals.
# NOT an exact atlas-label permutation. Requires many weakly dependent
# blocks, adequate mean specification, and calibration for the real grid.
wild_test <- function(D, H, H0, block, B=399L) {
  Q <- qr.Q(qr(H)); fitted <- Q %*% crossprod(Q,D)
  h <- rowSums(Q^2)
  U <- sweep(D-fitted, 1, sqrt(pmax(1-h, 1e-8)), "/")
  null_fit <- project(H0,D)
  statistic <- function(Z) {
    sum((project(H,Z)-project(H0,Z))^2)
  }
  observed <- statistic(D); g <- as.integer(factor(block))
  null <- replicate(B, {
    w <- sample(c(-1,1), max(g), replace=TRUE)
    statistic(null_fit + U*w[g])
  })
  c(statistic=observed, p_approx=(1+sum(null>=observed))/(B+1))
}
set.seed(19022026)
wild <- rbind(
```

```
  joint=wild_test(D,H,matrix(numeric(0),N,0),block),
  atlas=wild_test(D,H,E,block),
  interaction=wild_test(D,H,H[,1,drop=FALSE],block))
print(wild)
q3_pairs <- t(vapply(pairs,function(k) {
  Z <- (atlases[[k[2]]]-atlases[[k[1]]])/sqrt(2)
  M <- project(H[,1,drop=FALSE],Z); F <- project(H,Z)
  total <- sum(Z^2)
  c(atlas_fraction=sum(M^2)/total,
    spatial_fraction=sum((F-M)^2)/total)
},numeric(2)))
rownames(q3_pairs) <- pair_names

# 4. Local temporal dissimilarity with loss and gain components.
local_change <- function(X,Y) {
  a <- rowSums(X==1 & Y==1); b <- rowSums(X==1 & Y==0)
  c <- rowSums(X==0 & Y==1); den <- a+b+c
  data.frame(a=a,loss=b,gain=c,
    TBI=ifelse(den>0,(b+c)/den,NA_real_),
    loss_part=ifelse(den>0,b/den,NA_real_),
    gain_part=ifelse(den>0,c/den,NA_real_))
}

# 5. Spatial beta diversity. Double-empty comparisons are undefined.
jaccard_matrix <- function(X) {
  a <- tcrossprod(X); richness <- rowSums(X)
  union <- outer(richness,richness,"+")-a
  out <- 1-a/union
  diag(out) <- 0
  out
}
beta_mean <- function(X) {
  d <- jaccard_matrix(X)
  values <- d[lower.tri(d)]
  if (anyNA(values)) stop("Resolve empty-cell policy before spatial beta.")
  mean(values)
}
beta <- vapply(atlases,beta_mean,numeric(1))
print(beta)

# Paired spatial block bootstrap; all species and atlases move together.
# Intervals concern spatial-process generalisation, not detection error.
set.seed(19032026)
blocks <- split(seq_len(N),block)
Bboot <- 399L
boot_beta <- replicate(Bboot, {
  chosen <- sample.int(length(blocks),length(blocks),replace=TRUE)
  ii <- unlist(blocks[chosen],use.names=FALSE)
  vapply(atlases,function(X) beta_mean(X[ii,,drop=FALSE]),numeric(1))
})
beta_intervals <- t(vapply(pairs,function(k) {
  quantile(boot_beta[k[2],]-boot_beta[k[1],],c(.025,.975))
},numeric(2)))
rownames(beta_intervals) <- pair_names
print(beta_intervals)

# Baseline-fixed random allocation of observed losses and gains.
# This is our proposed conditional null, not the TBI package shuffle.
random_later <- function(X,Y,strata=rep(1,nrow(X))) {
  out <- X
  for (ii in split(seq_len(nrow(X)),strata)) {
    for (s in seq_len(ncol(X))) {
      on <- ii[X[ii,s]==1]; off <- ii[X[ii,s]==0]
      L <- sum(X[ii,s]==1 & Y[ii,s]==0)
```

```
      G <- sum(X[ii,s]==0 & Y[ii,s]==1)
      if (L>0) out[pick(on,L),s] <- 0L
      if (G>0) out[pick(off,G),s] <- 1L
    }
  }
  out
}
set.seed(19042026)
Bnull <- 399L                    # Teaching run, not final precision.
null_results <- vector("list",3)
for (j in seq_along(pairs)) {
  k <- pairs[[j]]; X <- atlases[[k[1]]]; Y <- atlases[[k[2]]]
  obs <- local_change(X,Y)
  if (anyNA(obs$TBI)) stop("Resolve double-empty temporal comparisons.")
  simulations <- replicate(Bnull, {
    Ystar <- random_later(X,Y)
    stopifnot(all(colSums(Ystar)==colSums(Y)))
    stopifnot(all(colSums(X==1 & Ystar==0)==colSums(X==1 & Y==0)),
              all(colSums(X==0 & Ystar==1)==colSums(X==0 & Y==1)))
    c(local_change(X,Ystar)$TBI, beta_mean(Ystar)-beta_mean(X))
  })
  obs$p_upper <- (1+rowSums(simulations[seq_len(N),]>=obs$TBI))/(Bnull+1)
  obs$null_mean <- rowMeans(simulations[seq_len(N),])
  obs$excess <- obs$TBI-obs$null_mean
  # Pool the three p vectors before any across-contrast adjustment.
  delta <- beta[k[2]]-beta[k[1]]
  null_delta <- simulations[N+1,]
  null_results[[j]] <- list(local=obs, null_delta=null_delta,
    observed_delta=delta,
    p_lower_than_null=(1+sum(null_delta<=delta))/(Bnull+1),
    p_higher_than_null=(1+sum(null_delta>=delta))/(Bnull+1))
}
# Holm adjustment covers all cells in all three separate contrasts.
# With only 399 replicates these adjusted tests have coarse resolution.
p_all <- unlist(lapply(null_results,function(z) z$local$p_upper))
p_holm <- matrix(p.adjust(p_all,method="holm"),nrow=N,ncol=3)
for (j in seq_along(pairs)) {
  null_results[[j]]$local$p_holm <- p_holm[,j]
  write.csv(cbind(xy,null_results[[j]]$local),
    paste0("atlas_outputs/local_",pair_names[j],".csv"),row.names=FALSE)
}
print(vapply(null_results,function(z) z$p_lower_than_null,numeric(1)))

# Base R plotting examples. The letter's figures use these saved data.
pdf("atlas_outputs/example_plots.pdf",width=9,height=6)
barplot(t(changes[,c("persist","loss","gain")]),beside=FALSE,
  col=c("grey60","#CC6677","#4477AA"),
  legend.text=c("Persistent presence","Loss","Gain"),
  main="Simulated complete-matrix change",ylab="Species-cell records")
par(mfrow=c(1,3))
for (t in 1:3) hist(occupancy[,t],breaks=seq(0,1,.1),
  xlim=c(0,1),ylim=c(0,10),main=names(atlases)[t],xlab="Proportion occupied")
par(mfrow=c(1,1))
plot(ecdf(occupancy[,1]),main="Simulated occupancy distributions",
     xlab="Proportion of cells occupied",ylab="Cumulative proportion of species")
lines(ecdf(occupancy[,2]),col=2); lines(ecdf(occupancy[,3]),col=4)
plot(occupancy[,1],occupancy[,3],xlim=c(0,1),ylim=c(0,1),
     xlab="Atlas 1 occupancy",ylab="Atlas 3 occupancy")
abline(0,1,lty=2)
local13 <- null_results[[3]]$local
par(mfrow=c(1,2))
image(1:12,1:8,matrix(local13$TBI,12,8),zlim=c(0,1),
  main="Simulated temporal Jaccard, 1 to 3",xlab="x",ylab="y")
plot(1:3,beta,type="b",xaxt="n",ylim=c(0,1),
```

```
  main="Simulated spatial beta",xlab="Atlas",ylab="Mean spatial Jaccard")
axis(1,at=1:3)
par(mfrow=c(1,1))
hist(null_results[[3]]$null_delta,main="Simulated conditional null, 1 to 3",
  xlab="Change in mean spatial Jaccard")
abline(v=beta[3]-beta[1],col=2,lwd=2)
dev.off()

# Save all objects needed for the document figures and numerical checks.
write.csv(changes,"atlas_outputs/matrix_changes.csv")
write.csv(occupancy,"atlas_outputs/occupancy.csv")
write.csv(range_change,"atlas_outputs/range_changes.csv")
write.csv(range_summary,"atlas_outputs/range_summary.csv")
write.csv(data.frame(atlas=names(beta),beta=beta),"atlas_outputs/spatial_beta.csv",
        row.names=FALSE)
write.csv(beta_intervals,"atlas_outputs/beta_intervals.csv")
write.csv(t(boot_beta),"atlas_outputs/boot_beta.csv",row.names=FALSE)
write.csv(data.frame(SS=SS),"atlas_outputs/q3_partition.csv")
write.csv(wild,"atlas_outputs/q3_wild.csv")
write.csv(q3_pairs,"atlas_outputs/q3_pairs.csv")
write.csv(sapply(null_results,function(z) z$null_delta),
        "atlas_outputs/null_beta.csv",row.names=FALSE)
for (t in 1:3) write.csv(cbind(xy,atlases[[t]]),
  paste0("atlas_outputs/atlas",t,".csv"),row.names=FALSE)
writeLines(capture.output(sessionInfo()),"atlas_outputs/session_info.txt")

# Optional established-package equivalents, documentation checked.
# vegan::vegdist(A1, method="jaccard", binary=TRUE)
# transport::wasserstein1d(occupancy[,1],occupancy[,2],p=1)
# betapart::beta.pair(as.data.frame(A1),index.family="jaccard")$beta.jac
# adespatial::TBI(A1,A2,method="jaccard",nperm=0,
#   test.BC=FALSE,test.t.perm=FALSE,save.BC=TRUE)
# For Hellinger sensitivity, transform EACH atlas separately:
# xs_h <- lapply(atlases,function(X) sqrt(X/rowSums(X)))
# D_h <- contrast_data(xs_h)
# Refit the same design to D_h; explain the changed weighting.
```

# References

Peer-reviewed methodological and empirical sources cited in the letter. Official software documentation is listed in Appendix A. Each DOI below is a clickable link.


Anderson, M. J. & ter Braak, C. J. F. (2003). Permutation tests for multi-factorial analysis of variance. Journal of Statistical Computation and Simulation, 73(2), 85–113. https://doi.org/10.1080/00949650215733

Anderson, M. J., Ellingsen, K. E. & McArdle, B. H. (2006). Multivariate dispersion as a measure of beta diversity. Ecology Letters, 9, 683–693. https://doi.org/10.1111/j.1461-0248.2006.00926.x

Baselga, A. (2010). Partitioning the turnover and nestedness components of beta diversity. Global Ecology and Biogeography, 19, 134–143. https://doi.org/10.1111/j.1466-8238.2009.00490.x

Baselga, A., Bonthoux, S. & Balent, G. (2015). Temporal beta diversity of bird assemblages in agricultural landscapes: land cover change vs. stochastic processes. PLOS ONE, 10(5), e0127913. https://doi.org/10.1371/journal.pone.0127913

Boakes, E. H., McGowan, P. J. K., Fuller, R. A., Ding Chang-qing, Clark, N. E., O'Connor, K. & Mace, G. M. (2010). Distorted views of biodiversity: spatial and temporal bias in species occurrence data. PLOS Biology, 8(6), e1000385. https://doi.org/10.1371/journal.pbio.1000385

Chase, J. M., Kraft, N. J. B., Smith, K. G., Vellend, M. & Inouye, B. D. (2011). Using null models to disentangle variation in community dissimilarity from variation in α-diversity. Ecosphere, 2(2), article 24. https://doi.org/10.1890/ES10-00117.1

Djogbenou, A. A., MacKinnon, J. G. & Nielsen, M. Ø. (2019). Asymptotic theory and wild bootstrap inference with clustered errors. Journal of Econometrics, 212(2), 393–412. https://doi.org/10.1016/j.jeconom.2019.04.035

Graves, G. R. & Rahbek, C. (2005). Source pool geometry and the assembly of continental avifaunas. Proceedings of the National Academy of Sciences of the USA, 102(22), 7871–7876. https://doi.org/10.1073/pnas.0500424102

Legendre, P. (2019). A temporal beta-diversity index to identify sites that have changed in exceptional ways in space–time surveys. Ecology and Evolution, 9, 3500–3514. https://doi.org/10.1002/ece3.4984

Legendre, P. & Gallagher, E. D. (2001). Ecologically meaningful transformations for ordination of species data. Oecologia, 129, 271–280. https://doi.org/10.1007/s004420100716

Legendre, P., De Cáceres, M. & Borcard, D. (2010). Community surveys through space and time: testing the space–time interaction in the absence of replication. Ecology, 91(1), 262–272. https://doi.org/10.1890/09-0199.1

Martínez-Camblor, P., Carleos, C. & Corral, N. (2014). Cramér-Von Mises statistic for repeated measures. Revista Colombiana de Estadística, 37(1), 45–67. https://doi.org/10.15446/rce.v37n1.44357

O'Sullivan, J. D., Terry, J. C. D. & Rossberg, A. G. (2023). Temporally robust occupancy frequency distributions in riverine metacommunities explained by local biodiversity regulation. Global Ecology and Biogeography, 32(12), 2230–2243. https://doi.org/10.1111/geb.13756

Suhonen, J. & Jokimäki, J. (2019). Temporally stable species occupancy frequency distribution and abundance–occupancy relationship patterns in urban wintering bird assemblages. Frontiers in Ecology and Evolution, 7, article 129. https://doi.org/10.3389/fevo.2019.00129

Terry, J. C. D., Langdon, W. & Rossberg, A. G. (2023). Codistribution as an indicator of whole metacommunity response to environmental change. Ecography, 2023, e06605. https://doi.org/10.1111/ecog.06605

Wayman, J. P., Sadler, J. P., Martin, T. E., Graham, L. J., White, H. J., Tobias, J. A. & Matthews, T. J. (2024). Unravelling the complexities of biotic homogenization and heterogenization in the British avifauna. Journal of Animal Ecology, 93, 1288–1302. https://doi.org/10.1111/1365-2656.14145